NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 67**

REVISION ASSINGMENT # 11 | (LIMITS) | TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. The value of $\tan\left(\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1+2r}{1+r^2(r+1)^2}\right)\right)$ is -

(A) 1 (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) none of these

2. If $f(x)=\frac{1}{x^2-17x+66}$, then $f\left(\frac{2}{x-2}\right)$ is discontinuous at

(A) $x=2,\frac{7}{3},\frac{25}{11}$ (B) $x=2,\frac{8}{3},\frac{24}{11}$ (C) $x=2,\frac{7}{3},\frac{24}{11}$ (D) no where in its domain

3. If $f(x)=\cot^{-1}\left(\frac{2x}{1-x^2}\right)$ and $g(x)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ then the value of $\lim\limits_{x\to t}\frac{f(x)-f(t)}{g(x)-g(t)}$, $(0<t<1)$ is -

(A) 1 (B) –1 (C) 2 (D) 1/2

4. Let $f(x)$ is a non-zero differentiable function $\forall\ x\in R$ & satisfying $f(x+y)=f(x-y)\ \forall\ x, y\in R$, then -

(A) $f(1)+f(0)=f'(0)$ (B) $f(1)-f(0)=f'(0)$

(C) $f(0)+f(1)=f(2)$ (D) $3f(2)=2f(3)$

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5. If $f(x)=\sqrt{x+\sqrt{x-\sqrt{x+\sqrt{x-\sqrt{x}.....\infty}}}}$, then value of $\lim\limits_{x\to1^+}\frac{f(x)-\left(\frac{\sqrt{x}+1}{2}\right)}{(x-1)}$

(A) 1 (B) 0 (C) $\frac{3}{8}$ (D) $\frac{3}{4}$

6. Let $f(x)=\begin{cases}a+(x-b)^2 & \text{for } |x-b|\le k\\ c+|x-b| & \text{for } |x-b|>k\end{cases}$, then the positive value of k so that f(x) becomes differentiable at x = b – k and x = b + k is-

(A) $\frac{3}{2}$ (B) 1/2 (C) 1 (D) 2

7. If $f(x)=\lim\limits_{n\to\infty}\frac{e^x+e^{-x}x^{2n}}{e^xx^{2n}+e^{-x}}$, then -

(A) $f(1^+)=e^2$ (B) $f(1^-)=\frac{1}{e^2}$ (C) $f(1)=e^2$ (D) $f(1^+)=\frac{1}{e^2}$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. Let a,b,c be positive numbers such that a + b + c = 15. Then the value of $\lim\limits_{x\to0}\left(\frac{a^x+b^x+c^x}{3}\right)^{1/x}$ can be -

(A) 5 (B) 4 (C) $\frac{3}{2}$ (D) 6

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9. Let [x] be the greatest integer function $f(x) = \frac{\sin\frac{1}{4}\pi[x]}{[x]}$ is -

(A) not continuous at any point (B) continuous at $\frac{3}{2}$

(C) discontinuous at 2 (D) differentiable at $\frac{4}{3}$

10. If $f(x) = \begin{cases} x^m \cos\left(x - \frac{1}{x}\right) & x > 0 \\ 0 & x = 0 \end{cases}$, then -

(A) $\forall$ m $\in (-\infty, 0]$, $f(x)$ is discontinuous at x = 0

(B) $\forall$ m $\in (0, \infty)$, $f(x)$ is differentiable at x = 0

(C) $\forall$ m $\in (1,\infty)$, $f(x)$ is differentiable at x = 0

(D) $\forall$ m $\in (0, 1]$, $f(x)$ is continuous but not differentiable at x = 0

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 11 to 12

Consider the following functions

$f(x) = \begin{cases} 1 - x, & 1 \le x \le 2 \\ 3 - x, & 2 < x \le 3 \end{cases}$

$g(x) = x^4 + x^2 + 1$

$h(x) = x^3$

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On the basis of above informations, answer the following questions :

11. The function $gof(x)$ is -

(A) discontinuous at x = 2 (B) continuous but not derivable at x = 2

(C) continuous and derivable at x = 2 (D) non derivable at more that one point in [1, 3]

12. The function h o $f(x)$ is -

(A) discontinuous at x = 2 (B) continuous but not derivable at x = 2

(C) continuous and derivable at x = 2 (D) increasing in (1, 2)

Space for Rough Work

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SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. The function max.$\{1-x, 1+x, 2x^2\}$; $x \in R$ is discontinuous at 'p' points and non-differentiable at q points then p + q is

2. If $f : R \to R^+$ is differentiable function such that $|f(y).f(x-y)|+(x-y)^2 \geq |f(x).f(x-y)| \forall x, y \in R$, $f(0) = 5$, then the value of $f(5)$ is

3. If $f(x) = \begin{cases} \tan^{-1}\left(\frac{ax+b}{c}\right) & 0 < |x| < \frac{1}{2} \\ 0 & x = \frac{1}{2} \\ \log\left(2+bx^2\right) & \frac{1}{2} < x < \frac{1}{\sqrt{2}} \end{cases}$ is differentiable in $x \in \left(0, \frac{1}{\sqrt{2}}\right)$, then $\left|\frac{a+b}{c}\right|$ is equal to

SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I	**Column-II**
(A)	The number of the values of x in $(0, 2\pi)$, where the function $f(x) = \frac{\tan x + \cot x}{2} - \left\|\frac{\tan x - \cot x}{2}\right\|$ is continuous but not differentiable is	(p) 2
(B)	The number of points where the function $f(x) = \min\{1, 1+x^3, x^2-3x+3\}$ is non-derivable	(q) 0
(C)	The number of points where $f(x) = (x+4)^{1/3}$ is non-differentiable is	(r) 4
(D)	Consider $f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{x.2}{\pi}\right)+\frac{\pi}{2}, & 0 < x \leq \frac{\pi}{2} \\ \sin^{-1}\sin x, & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$ Number of points in $\left(0, \frac{3\pi}{2}\right)$, where $f(x)$ is non-differentiable is	(s) 1

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2.

	Column - I		**Column - II**
(A)	Number of points where the function $f(x) = \begin{cases} 1+\left[\cos\frac{\pi x}{2}\right] & 1<x\le 2 \\ 1-\{x\} & 0\le x<1 \\ \lvert\sin \pi x\rvert & -1\le x<0 \end{cases}$ and $f(1) = 0$ is continuous but non-differentiable where [ ] denote greatest integer and { } denote fractional part function	(p)	0
(B)	$f(x) = \begin{cases} x^2 e^{1/x}, & x \neq 0 \\ 0 & , \ x = 0 \end{cases}$, then $f'(0^-)$ is equal to	(q)	1
(C)	The number of points at which $g(x) = \frac{1}{1+\frac{2}{f(x)}}$ is not differentiable where $f(x) = \frac{1}{1+\frac{1}{x}}$, is	(r)	2
(D)	Number of points where tangent does not exist for the curve $y = \text{sgn}(x^2 - 1)$	(s)	3

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REVISION ASSINGMENT # 10 FUNCTION & ITF MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	C	C	B	B	C	B	A,C,D	A,C,D	A,B,D
	Q.	11	12	13	14						
	A.	A,D	B	C	C						
SECTION-III	Q.	1	2	3	4						
	A.	2	2	7	2						

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